

Install

Set up R and Python to follow along live

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If you would like to run the demos on your laptop, install one or both of the `combss` packages below. The same demos exist in R and Python with matching results, so either language is fine.

R

```
install.packages("combss")
```

That's it — `combss` is on CRAN. The only run-time dependency is `glmnet`, which CRAN will install automatically. Tested with R 4.3 and 4.4 on macOS, Linux, and Windows.

Quick check

```
library(combss)
set.seed(1)
x <- matrix(rnorm(100 * 20), 100, 20)
y <- as.numeric(x %*% c(2, -1, 1, rep(0, 17))) + rnorm(100) * 0.3)
fit <- combss(x, y, family = "gaussian", q = 5)
fit$subset_list[1:5]
```

Expected output: nested supports building up to $\{1, 2, 3\}$ at $k = 3$.

Python

```
pip install combss
```

Python 3.9 or newer. Dependencies (`numpy`, `scipy`, `scikit-learn`, `pandas`) install automatically.

Quick check

```
import numpy as np
from combss import combss

rng = np.random.default_rng(1)
X = rng.standard_normal((100, 20))
beta = np.r_[2, -1, 1, np.zeros(17)]
y = X @ beta + 0.3 * rng.standard_normal(100)
fit = combss(X, y, family="gaussian", q=5)
print(fit.subset_list[:5])
```

Expected output: nested supports building up to $\{1, 2, 3\}$ at $k = 3$.

Editor recommendations

Either of these makes running the demos chunk-by-chunk smooth:

- **RStudio** — open the `.qmd` file, click the green  on each chunk.
- **VS Code** with the *Quarto* and *Jupyter* extensions — same idea, language-aware.

Cloning the demo repository

```
git clone https://github.com/saratmoka/combss-statfest.git
cd combss-statfest
```

All the data (Khan SRBCT and the simulation seeds) lives under `data/`. No external downloads are needed at run time.

Trouble?

- R cannot find the package: make sure you are on R 4.0 and check `available.packages()` lists `combss`. If not, your CRAN mirror may be stale; pick another with `chooseCRANmirror()`.
- Python `pip install` fails on `combss`: confirm you are on Python 3.9 with `python --version`; older versions are unsupported.
- Anything else: open an issue at <https://github.com/saratmoka/combss-statfest/issues>.